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FAST DISSOLVING TABLETS: FORMULATION APPROACHES, EMERGING TECHNOLOGIES, AND THERAPEUTIC APPLICATIONS

Rashmi Singh - Research Scholar | rrashmisingh2002@gmail.com

Department of Pharmacy, Kalinga University, Raipur, Chhattisgarh

Deepesh Kumar - Assistant Professor | deepesh.kumar@kalingauniversity.ac.in

Faculty of Pharmacy, Kalinga University, Raipur, Chhattisgarh

ABSTRACT:

Fast dissolving tablets (FDTs), also known as orally disintegrating tablets, dissolve rapidly in the mouth without water. They are particularly beneficial for paediatric, geriatric, and dysphagic patients who have difficulty swallowing conventional tablets. Formulated with superdisintegrants and porous structures, FDTs enable rapid disintegration and drug release, potentially improving patient convenience and treatment effectiveness. Various manufacturing techniques, including direct compression, freeze drying, moulding, spray drying, and sublimation, have been developed to enhance their performance and stability. This review discusses formulation strategies, manufacturing methods, advantages, limitations, therapeutic applications, marketed products, and recent technological advancements in FDTs.

Keywords: *Orally disintegrating tablets, Fast melt tablets, Patient-friendly dosage form, Rapid drug release, Super disintegrants, Taste masking techniques, Solid oral formulations, Novel drug delivery systems*

1. INTRODUCTION

Fast dissolving tablets (FDTs) are an advanced oral dosage form developed to improve ease of drug administration and patient convenience. These tablets quickly disperse in the mouth after placement on the tongue, eliminating the need for water and making them suitable for a wide range of patients [1,2]. Their simple method of use and rapid action have made them an important focus in pharmaceutical formulation research.

The oral route is commonly used for drug delivery because it is convenient, safe, and well accepted by patients [3]. However, conventional tablets and capsules may not be suitable for all individuals. Many patients, especially children, elderly people, and those with medical conditions affecting swallowing, face difficulty in taking solid dosage forms [4,5]. This can lead to poor compliance and reduced effectiveness of therapy.

FDTs were introduced to overcome these limitations. Once placed in the oral cavity, they break down quickly in saliva and release the drug for absorption without the need for chewing [6]. The disintegration process usually occurs within a short time, depending on the formulation and preparation method [7]. This rapid breakdown improves patient acceptability and may also result in a quicker therapeutic response.

The formulation of FDTs includes the use of super disintegrants and water-soluble excipients that promote fast tablet dispersion [8]. In certain cases, partial drug absorption may occur in the oral region before reaching the gastrointestinal tract, which can enhance bioavailability [9]. Various manufacturing methods such as direct compression, freeze-drying, moulding, spray drying, and sublimation are used to produce tablets with rapid disintegration and adequate strength [10,11,25].

Recent developments have focused on improving taste masking, stability, and overall patient experience. Due to these advantages, FDTs are widely accepted as a patient-friendly dosage form, particularly for individuals with swallowing difficulties. Continuous research in this area is further expanding their applications in modern pharmaceutical science [12].

2. OVERVIEW OF FAST DISSOLVING TABLETS

Fast dissolving tablets (FDTs) are oral solid dosage forms that quickly disintegrate in the mouth without water. They release the drug within seconds, which is then swallowed or partly absorbed through the oral mucosa [13,14,32]. Their fast action and ease of use make them an effective drug delivery option.

2.1 Definition and Basic Concept: FDTs dissolve rapidly in saliva, allowing drug intake without water or chewing. This makes them suitable for patients with swallowing difficulties and improves compliance [15].

2.2 Mechanism of Rapid Disintegration: Rapid disintegration is achieved using super disintegrants, water-soluble excipients, and porous structures. Contact with saliva causes swelling and tablet breakdown, leading to quick drug release [16].

2.3 Patient Acceptability and Compliance: FDTs should provide a smooth mouth feel without irritation [17,55]. Taste masking is important due to bitter drugs, so sweeteners and flavours are added [18]. These tablets improve convenience and overcome limitations of conventional dosage forms [19].

3. KEY PROPERTIES OF FAST DISSOLVING TABLETS

Fast dissolving tablets (FDTs) require suitable physicochemical and mechanical properties to ensure quick disintegration, stability, and good patient acceptability [20].

3.1 Mouth Feel: Mouth feel is important as FDTs disintegrate in the oral cavity. The tablet should produce a smooth and non-irritating sensation. Large or insoluble particles may cause discomfort, so particle size is controlled and flavouring agents are added to improve palatability [21,47].

3.2 Hygroscopicity: FDTs often contain porous and hydrophilic materials, making them sensitive to moisture. Moisture absorption can affect stability and performance, so protective packaging is required to prevent humidity exposure [22,53].

3.3 Mechanical Strength and Friability: These tablets may have lower strength due to their porous nature. High friability can cause breakage during handling. Therefore, formulation must balance rapid disintegration with sufficient mechanical strength [23,24,9,50].

4. ADVANTAGES AND LIMITATIONS OF FAST DISSOLVING TABLETS

Fast dissolving tablets (FDTs) improve drug administration by offering quick disintegration and better patient compliance, especially for individuals who have difficulty swallowing conventional dosage forms [12,15,18].

4.1 Advantages:

- **Better compliance:** Suitable for paediatric, geriatric, and dysphagic patients.
- **Fast action:** Rapid disintegration leads to quicker drug release and effect.
- **No water needed:** Easy to use in any situation.
- **Good acceptability:** Taste masking and flavouring improve palatability.
- **Accurate dosing:** More stable and precise than liquid forms [25,31,40].

4.2 Limitations:

- **Moisture sensitivity:** May affect stability due to hygroscopic nature.
- **Low strength:** Higher risk of breakage during handling.
- **Taste issues:** Bitter drugs require effective masking.
- **Limited drug load:** Not suitable for high-dose drugs [28,32,36].

5. SUITABLE DRUG CANDIDATES FOR FAST DISSOLVING TABLETS

FDTs are not suitable for all drugs. Selection of an appropriate drug is essential to ensure rapid disintegration, dissolution, and absorption [21,25].

5.1 Physicochemical Characteristics: Ideal drug candidates should have low to moderate dose, good water solubility, and acceptable taste [23,31]. High-dose drugs are usually unsuitable due to large tablet size. Bitter drugs require effective taste masking. In some cases, drugs with first-pass metabolism may benefit from partial absorption in the oral cavity, improving bioavailability [19,24,28,33,34].

5.2 Examples of Suitable Drugs:

Common FDT drugs include:

- **Antihistamines:** loratadine, cetirizine
- **Analgesics:** ibuprofen, paracetamol
- **Antiemetics:** ondansetron, domperidone
- **CNS drugs:** risperidone, olanzapine [27,33]

These drugs are preferred due to their need for rapid action and better patient compliance [30].

6. CHALLENGES IN FORMULATION DEVELOPMENT OF FAST DISSOLVING TABLETS

Despite their advantages, FDTs present several formulation challenges due to the need for rapid disintegration along with adequate strength, stability, and taste.

6.1 Taste Masking: Many drugs are bitter, so effective taste masking is essential. Common methods include coating, microencapsulation, and use of sweeteners or flavours.

6.2 Mechanical Strength: Low compression and porous structure may reduce hardness and increase friability, leading to possible tablet breakage during handling.

6.3 Moisture Sensitivity: Hydrophilic excipients make FDTs prone to moisture absorption, which can affect stability. Proper packaging is required to protect from humidity.

6.4 Drug Loading: High-dose drugs are difficult to formulate due to increased tablet size and reduced acceptability. Hence, FDTs are better suited for low to moderate doses.

6.5 Excipients Selection: Selection of suitable excipients is important to balance fast disintegration with sufficient strength and stability [37,39,41,48,52].

7. THERAPEUTIC APPLICATIONS OF FAST DISSOLVING TABLETS

Fast dissolving tablets (FDTs) are widely used due to their rapid action, ease of administration, and improved patient compliance. They are especially useful in conditions requiring quick relief or in patients with swallowing difficulty.

7.1 Allergic Disorders: Antihistamines like loratadine and cetirizine are used in FDTs for rapid relief of allergy symptoms.

7.2 Pain and Inflammation: Drugs such as ibuprofen, paracetamol, and diclofenac provide faster pain relief when formulated as FDTs.

7.3 Nausea and Vomiting: Antiemetics like ondansetron and domperidone are useful in FDT form for patients unable to swallow tablets.

7.4 CNS Disorders: Used in conditions like schizophrenia, Parkinson's disease, and migraine. Drugs include risperidone, olanzapine, and zolmitriptan.

7.5 Gastrointestinal Disorders: Drugs like famotidine, ranitidine, and Mosa pride help in quick relief from acid-related disorders.

8. MANUFACTURING TECHNIQUES OF FAST DISSOLVING TABLETS

Fast dissolving tablets are prepared using different conventional and advanced techniques to achieve rapid disintegration with sufficient mechanical strength. The selection of method depends on drug properties, desired performance, and manufacturing feasibility **[67,69,70]**.

8.1 Freeze Drying (Lyophilization): This method involves freezing a drug solution followed by drying under vacuum to create highly porous tablets. These tablets disintegrate very rapidly but are fragile and require special packaging **[65–67]**.

8.2 Moulding Technique: In this technique, the drug is moulded into tablets using a water-soluble matrix and low pressure. The tablets dissolve quickly but have comparatively lower strength **[60–62]**.

8.3 Sublimation Technique: Volatile substances are added and later removed to create pores in the tablet. This enhances saliva penetration and results in fast disintegration **[5,63–65]**.

8.4 Spray Drying: Spray drying produces porous powders that are later compressed into tablets. These tablets show rapid disintegration due to their porous nature **[66–68]**.

8.5 Mass Extrusion: This method is mainly used for taste masking. The drug is extruded into small segments and dried before formulation, ensuring uniform drug distribution **[51,64–66]**.

8.6 Direct Compression: It is a simple and cost-effective method where the drug and excipients are directly compressed. It is suitable for large-scale production but requires proper formulation design **[28,69,70]**.

9. ADVANCED AND PATENTED TECHNOLOGIES

Advanced technologies are used to improve fast dissolving tablets in terms of rapid disintegration, taste masking, mechanical strength, and patient acceptability. These systems help overcome limitations of conventional methods **[57,64,69]**.

9.1 Zydis Technology: Zydis technology uses freeze-drying to produce highly porous tablets that disintegrate very quickly in saliva. The formulation contains polymers and sugars for stability and taste improvement. However, these tablets are fragile and require special packaging **[57,64]**.

9.2 OraSolv Technology: OraSolv uses taste-masked drug particles along with effervescent agents to promote rapid disintegration. Tablets are prepared using low compression pressure, which maintains fast dissolution. However, they may have lower mechanical strength and need careful handling **[35,57]**.

9.3 DuraSolv Technology: DuraSolv is an improved version that provides better mechanical strength compared to OraSolv. Tablets are manufactured using conventional equipment and can be packed easily. It is mainly suitable for low-dose drugs **[48,52]**.

9.4 WOWTAB Technology: WOWTAB uses a combination of saccharides with different moulding properties to produce tablets. This approach provides both good mechanical strength and rapid disintegration. It also improves mouth feel and patient acceptability **[63,69]**.

9.5 FlashDose Technology: FlashDose technology forms a floss-like porous matrix using flash heat processing. This structure allows rapid penetration of saliva, leading to quick tablet disintegration. It is widely used for fast-acting formulations **[41,57]**.

9.6 FlashTab Technology: FlashTab involves the use of micro granulated drug particles that are compressed into tablets. These tablets provide rapid disintegration while maintaining good strength and stability. It also ensures uniform drug distribution **[59,70]**.

9.7 OraQuick Technology: OraQuick uses a microsphere-based taste-masking system to hide unpleasant drug taste. It does not require organic solvents, making it safer and efficient. Tablets prepared show good strength and fast disintegration **[52,66]**.

Overall, these technologies improve the performance and acceptability of FDTs **[57,69]**.

10. MECHANISM OF TABLET DISINTEGRATION

Fast dissolving tablets disintegrate due to the action of super disintegrants when they come in contact with saliva. Usually, more than one mechanism works together to achieve rapid tablet breakdown **[35,57,69]**.

10.1 Capillary Action (Wicking): In this mechanism, saliva enters the tablet through pores by capillary action. This replaces air inside the tablet and weakens particle bonds, leading to disintegration. It is one of the most important mechanisms **[48,63]**.

10.2 Swelling: Certain disintegrants absorb water and swell in size when in contact with saliva. This swelling creates internal pressure, which breaks the tablet into smaller particles. It significantly speeds up disintegration **[52,66]**.

10.3 Heat of Wetting: Some excipients produce heat when they come in contact with water. This heat increases water penetration and weakens the tablet structure. Although less common, it supports other mechanisms **[41,57]**.

10.4 Gas Generation: Effervescent agents react with saliva to produce carbon dioxide gas. The generated gas creates pressure inside the tablet, causing it to break apart quickly. This method enhances rapid dispersion **[59,69]**.

10.5 Enzymatic Action: Salivary enzymes may break down certain excipients present in the tablet. This weakens the structure of the tablet and helps in its disintegration. It plays a supportive role in some formulations **[63,70]**.

10.6 Particle Repulsion: Water may create repulsive forces between particles inside the tablet. These forces reduce attraction between particles, leading to their separation. This results in faster tablet breakup **[52,70]**.

10.7 Deformation Recovery: During compression, some particles get deformed. When exposed to water, they return to their original shape, creating stress within the tablet. This stress helps in breaking the tablet structure **[57,69]**.

11. MARKETED PRODUCTS OF FAST DISSOLVING TABLETS

Fast dissolving tablets (FDTs) are widely commercialized by many pharmaceutical companies due to their rapid action and patient convenience. These products are available for different therapeutic uses such as allergy, pain, gastrointestinal disorders, and nausea. Their market presence shows the growing importance of FDT technology in modern drug delivery **[63,69]**.

Table 1: Marketed FDT Products

Product Name	Active Drug	Manufacturer	Country	Remark
Imodium Lingual	Loperamide	R. P. Scherer Corp.	USA	Used for diarrhoea
Pepcid Rapitab	Famotidine	Merck & Co.	USA	Anti-ulcer drug
Mosid-MT	Mosa pride Citrate	Torrent Pharma	India	GI disorder treatment
Claritin RediTabs	Loratadine	Schering-Plough	USA	Antihistamine
Nimulid-MD	Nimesulide	Panacea Biotech	India	Anti-inflammatory
Zyrof Meltab	Rofecoxib	Zydus Cadila	India	Pain relief
Zofran ODT	Ondansetron	GlaxoSmithKline	UK	Antiemetic
Maxalt-MLT	Rizatriptan	Merck & Co.	USA	Migraine treatment

These examples show the wide clinical use and commercial success of FDTs **[63,69]**.

12. PACKAGING CONSIDERATIONS

Packaging is very important for maintaining the stability of FDTs, as they are porous and sensitive to moisture. Exposure to humidity can affect tablet integrity and performance, so protective systems are required **[57,69]**.

FDTs are usually fragile, so packaging must also prevent mechanical damage during handling and transport. Aluminium blister packs or alu–alu packaging are commonly used to provide moisture protection and strength **[63,70]**.

In some cases, unit-dose packaging like peelable blisters or sachets is used for better safety and convenience. Proper packaging ensures product quality, stability, and effectiveness throughout shelf life **[52,69]**.

13. FUTURE PERSPECTIVES AND CONCLUSION

Fast dissolving tablets have become an important oral drug delivery system due to their rapid action and ease of use. They are especially beneficial for paediatric, geriatric, and dysphagic patients, improving compliance and treatment outcomes.

Significant progress has been made in formulation techniques and technologies to enhance disintegration, strength, and patient acceptability. The availability of many marketed products highlights their growing importance in the pharmaceutical industry.

In the future, research will focus on better taste masking, improved mechanical strength, and expanding drug compatibility. Advances in excipients and drug delivery systems are expected to further improve FDT performance.

In conclusion, FDTs are a promising and patient-friendly dosage form with strong potential for future pharmaceutical development **[63,69]**.

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